# The LQ Digital Mode Family: Protocol Architecture and Reference Specification for Weak-Signal Communications

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***Abstract*—This paper introduces LQ8, an open-source weak-signal digital mode for amateur radio that completes structured two-way contacts in 4 transmissions (60 s), achieving a 1.50× speedup over canonical 6-transmission exchanges (90 s) and 1.25× over 5-transmission `RR73` exchanges (75 s) while preserving complete bidirectional exchange of Maidenhead grid locators, signal reports, and mutual acknowledgments. By replacing fixed-width message-type headers with variable-length prefix codes matched to physical-layer bit budgets, LQ8 packs the answering station's grid locator and measured signal report into a single 77-bit reply. In multi-station pileup operations, LQ8 confirms up to two answering stations simultaneously within a single-carrier 50.0 Hz transmission with a 0.0 dB RF power-splitting penalty, achieving a peak rate of 240 QSOs/h in continuous pileup (4× over single-carrier FT8; 480 QSOs/h dual-carrier) while retaining −21.0 dB SNR sensitivity. The protocol family also defines three extended profiles (LQ16, LQ4, and LQ2) tailored for extreme sensitivity, VHF/UHF contesting, and fast-burst channels.**

## I. INTRODUCTION

The FT8 digital mode [1], designed by Steve Franke (K9AN) and Joe Taylor (K1JT), has transformed amateur radio since its introduction in 2017 in the WSJT-X software suite [2], enabling reliable contacts at signal-to-noise ratios as low as −21.0 dB in a 2500 Hz bandwidth. Its 77-bit message payload, combined with a low-density parity-check (LDPC) forward error correction code [5] and 8-tone Gaussian frequency-shift keying (8-GFSK), provides a robust physical layer that is now well-proven across millions of contacts worldwide.

In conventional weak-signal protocols (e.g., FT8), a structured two-way contact canonically requires six sequential transmissions across 15-second time slots (90 seconds total), or five transmissions (75 seconds) when using shortened `RR73` confirmations (CQ → CALL → REPORT → RREPORT → RR73). Under a 77-bit payload budget, encoding two 28-bit standard callsigns, a 15-bit Maidenhead grid locator, and a 5-bit signal report requires 76 information bits ($28+28+15+5=76$). Uniform fixed-width type headers (typically 3 or 4 bits) exceed the remaining 1-bit margin, preventing the simultaneous transmission of locator and report in a single transmission.

**LQ8** resolves this constraint by employing **variable-length prefix codes** [4] tailored to physical-layer bit budgets. By allocating a 1-bit prefix to the dominant response message format, LQ8 preserves 76 bits for payload data, packing both the answering station's grid locator and measured signal report into a single frame. This completes confirmed contacts in 4 transmissions (60 seconds) with full mutual verification, while retaining the proven continuous-phase 8-GFSK and LDPC$(174,91)$ physical layer (−21.0 dB SNR sensitivity). Furthermore, for high-rate DXpedition Fox and Hound (F/H) pileups, LQ8 introduces the MULTI-REPORT+73 protocol to confirm two answering Hound stations simultaneously within a single-carrier transmission without RF power-splitting penalties. Cross-isolated Costas synchronization arrays ($C_7=[2,5,6,1,3,0,4]$, exhibiting $\leq 1$ coincidence hit) isolate traffic from legacy decoders. Physical layer, CRC-14, and LDPC parameters are presented in Section V.

Building upon LQ8, the protocol family defines three complementary profiles (LQ16, LQ4, and LQ2) tailored for ultra-weak fading channels, rapid contesting, and fast burst links.

### A. Architectural Philosophy & Design Principles

The LQ family is engineered around four core design tenets:

- **Payload-Matched Prefix Allocation:** Replacing uniform headers with variable-length prefix codes frees capacity for simultaneous locator and report transmission, shortening 1-on-1 contacts to 4 slots (60 s) and reducing vulnerability to ionospheric fading (QSB) and transient interference.
- **Coexistence and Spectrum Preservation:** All modes maintain continuous-phase GFSK and LDPC$(174,91)$ channel coding while employing low-cross-coincidence Costas sync arrays to prevent cross-protocol false decodes.
- **Robust Special-Case Support:** Non-standard callsigns (up to 13 characters), dedicated 1-bit portable suffixes (`0`=*none*, `1`=`/P`), 24/20-bit collision-resistant callsign hashes, and a 104-character free-text Varicode provide seamless support for contesting, portable activations (POTA/SOTA), and DXpeditions.
- **Clear-Text Station Identification:** Transmitting stations convey their callsigns in clear text across all 1-on-1 exchanges (including non-standard caller replies via 48-bit Base-38 encoding), reserving compact hashes strictly for target addressing and multi-station pileup frames.

### B. Implementations & Reference Software

The complete suite is implemented in the open-source C++ reference library (`lq_lib`) [11] (MIT License), featuring an automated verification harness and golden vector test suites across all four modulation profiles. Web demonstrations and interactive protocol resources are published on the official project portal [12]. Native portable operation is provided by **qFT8** [13], deploying LQ8 alongside live testbeds for LQ16, LQ4, and LQ2.

## II. QSO PROTOCOL

The LQ protocol establishes deterministic state transitions, transmission slot timing, and multi-station arbitration to achieve confirmed bilateral contacts across low-SNR channels.

### A. Standard Four-Message Exchange

The LQ QSO protocol is built upon two core principles: bidirectional exchange of Maidenhead grid locators and signal reports (measured SNR in dB), coupled with mutual confirmation (Table 1). Following weak-signal convention, LQ establishes an operational termination criterion: each station must hear an explicit confirmation after delivering its own signal report before considering the contact complete. Station A considers the contact complete upon receiving Station B's concluding 73 in T4, while Station B considers it complete upon decoding Station A's report in T3 and transmitting its final 73.

**Table 1:** QSO sequence comparison between conventional protocols (FT8) and the LQ8 4-slot protocol.

| Seq | Conventional Protocols (FT8) | Seq | LQ8 Protocol |
|---|---|---|---|
| 1 | `CQ YO1YO JN47` | 1 | `CQ YO1YO JN47` |
| 2 | `YO1YO TU2TU KL22` | 2 | `YO1YO TU2TU KL22 -03` |
| 3 | `TU2TU YO1YO +05` | 3 | `TU2TU YO1YO R+05` |
| 4 | `YO1YO TU2TU R-03` | 4 | `YO1YO TU2TU 73` *[Confirmed]* |
| 5 | `TU2TU YO1YO RR73` (or `RRR`) | — | — |
| 6 | `YO1YO TU2TU 73` (optional if `RR73`) | — | — |

As summarized in Table 1, LQ8 completes a contact in 4 time slots (60 s vs. 5–6 messages / 75–90 s in conventional exchanges ending in `RR73` or `RRR`/73), delivering full bidirectional callsigns, Maidenhead locators, and signal reports with mutual verification:

1. **T1 (Slot 1, Even):** Station A transmits a CQ broadcast (catalogued in Section III, Types 1–4) containing its callsign, optional portable suffix, grid locator, or CQ modifier.
2. **T2 (Slot 2, Odd):** Station B transmits a directed CALL reply (Types 5–7) packing Station A's target address, caller callsign, measured SNR report, and grid locator into a single frame.
3. **T3 (Slot 3, Even):** Station A acknowledges with REPORT+73 (Type 8 for standard callsigns, or multi-station Type 11), delivering its measured SNR report for Station B.
4. **T4 (Slot 4, Odd):** Station B transmits a final 73 acknowledgment (Types 9 or 10; Type 12 for 10–13-character compound callsigns), completing the confirmed bidirectional contact.

### B. Retransmission, Error Recovery and Logging Semantics

To ensure session progress over lossy channels, LQ formalizes timeout recovery and logging rules.

**Retransmission and Repeat Calls.** If Station B misses REPORT+73 in T3 (due to QSB or interference), it repeats CALL in the next odd slot. Receiving this repeat informs Station A that its report was lost, prompting Station A to resend REPORT+73. Similarly, if Station A misses the final 73 in T4, it retransmits REPORT+73; Station B reacknowledges with 73 without duplicate logging.

**QSO Logging Semantics.** Contacts are logged once callsigns, locators, reports, and acknowledgments are exchanged: Station A logs upon decoding the final 73 in T4 (or transmitting REPORT+73 in Fox pileup mode); Station B logs upon transmitting 73 in T4 (having decoded Station A's report in T3). In ADIF logging (LoTW, QRZ.com, Club Log), contacts map to mode `DATA` with submode `LQ8` (`LQ4`, `LQ2`, `LQ16`). A watchdog aborts after three unacknowledged attempts (45 s in LQ8, 22.5 s in LQ4), returning to idle monitoring.

### C. DXpedition Fox and Hound Multi-Station Pileup Operation

During DXpedition pileups or contest operations in Fox and Hound (F/H) mode, Station A (Fox) activates the MULTI-REPORT+73 protocol (Type 11) to confirm up to two answering Hound stations simultaneously on a single carrier:

1. **Slot 1 (Fox):** Fox broadcasts directed CQ on its primary run frequency (e.g., `CQ DX FOX`).
2. **Slot 2 (Hounds):** Remote Hounds call simultaneously on split audio frequencies via directed CALL (e.g., `<FOX> <HOUND1> FN42 -08`).
3. **Slot 3 (Fox):** Fox confirms both Hounds simultaneously on the run frequency via MULTI-REPORT+73 (`<HOUND1> R+05 <HOUND2> R-02 <FOX>`).
4. **Slot 4 (Hounds):** Both Hounds respond with concluding 73 acknowledgments on their split frequencies (`<FOX> <HOUND1> 73`).

When confirming a single Hound in Fox mode, standard callers are confirmed using Type 8 (REPORT+73 `STD+SUF`). Single-target Type 11 (or Type 12 for final acknowledgments) is used when confirming non-standard callers or when operating under strict split-frequency F/H scheduling; in this case, Target 2 duplicates Target 1 (identical hash, and identical SNR in Type 11), which decoders process as a single-station transmission.

**Throughput Multiplier Analysis.** In continuous pileups where Step 3 overlaps with incoming calls, completing 2 QSOs every 2 slots (30 s) delivers a peak rate of 240 QSOs/h (4× over single-carrier FT8; 480 QSOs/h dual-carrier; 160 QSOs/h in isolated 3-slot exchanges) with **0.0 dB power penalty**, avoiding the −3.0 dB RF loss of multi-carrier FT8 while halving fading exposure.

### D. Multi-Caller Contention and Pileup Queuing

When multiple stations answer the same CQ simultaneously in 1-on-1 mode, Station A selects one caller and transmits REPORT+73 addressed to that station; other callers retry in the next alternating time slot. Frames are disambiguated via uniform 24-bit callsign hashes ($2^{24} \approx 16.78 \times 10^6$ bins, Section IV-D) and canonical precedence (Section III-C).

To optimize pileup resolution, transceivers implement opportunistic candidate queuing (up to 16 stations). Priority aging balances throughput with fairness, preventing high-SNR arrivals from starving marginal callers. Upon decoding 73 in T4, Station A immediately directs REPORT+73 to the highest-priority queued station without an intermediate CQ, accelerating turnover. If the queue is full, lowest-SNR entries below decode threshold are evicted first; entries expire after four unacknowledged cycles (60 s in LQ8, 30 s in LQ4).

## III. MESSAGE ARCHITECTURE

To maximize channel efficiency within the 77-bit physical frame budget, LQ replaces fixed-width bit fields with an entropy-matched variable-length message architecture.

### A. Design Motivation & Prefix-Free Framing

Under a strict 77-bit physical payload budget, conventional fixed-width message headers impose an unnecessary penalty on high-entropy transmissions. In a standard CALL reply, encoding two 28-bit standard callsigns, a 15-bit Maidenhead grid locator, and a 5-bit measured SNR report requires 76 information bits:

$$\underbrace{1\text{b}}_{\text{Prefix}} + \underbrace{28\text{b}}_{\text{Target}} + \underbrace{28\text{b}}_{\text{Caller}} + \underbrace{15\text{b}}_{\text{Locator}} + \underbrace{5\text{b}}_{\text{SNR}} = \mathbf{77}\ \text{bits}. \quad (1)$$

Uniform 3-bit or 4-bit type headers exceed the remaining 1-bit margin, preventing simultaneous transmission of locator and report in conventional protocols.

LQ resolves this constraint by synthesizing variable-length prefix coding [4, 3] with physical-layer bit budgets. Assigning a **1-bit** codeword (`1`) to the dominant CALL STD (NO-SUF) format preserves 76 bits for payload data, combining station identification, locator delivery, and signal report measurement into a single frame. Simpler messages (such as standard CQ, requiring 43 payload bits) readily accommodate longer prefix codes.

Formally, the prefix code $\mathscr{C} = \{c_1, c_2, \ldots, c_k\}$ is *prefix-free*:

$$\forall i \neq j: \quad c_i \text{ is not a prefix of } c_j. \quad (2)$$

The receiver traverses a binary prefix tree bit-by-bit from the most-significant bit until reaching a terminal leaf node, identifying the message type without length fields or delimiters. Although the mathematical binary prefix tree across all 16 slots satisfies $\sum_{i=1}^{16} 2^{-\text{len}_i} = 1.0$, the active operational codebook is intentionally prefix-free without being Kraft-complete because Types 14–16 are explicitly reserved for future backward-compatible protocol expansions rather than active operational message formats.

### B. Operational Categories & Message Catalogue

Stations select transmission formats according to the operational categories defined in Table 2. Table 3 catalogues all 16 message slots with their prefix codewords, payload layouts, and bit budgets. Frame payloads $\leq 77$ bits are zero-padded to the 77-bit link-layer boundary. Payloads are serialized in big-endian bit order (MSB first) into a 10-byte buffer: bit 0 corresponds to `0x80` of Byte 0, bit 7 to `0x01` of Byte 0, and bits 77–79 (the three least significant bits of Byte 9) are set to zero.

**Field Designations in Table 3:**

- **Canonical Short Identifiers:** Standardized mnemonics: **CQ**, **CALL**, **RPT73** (for REPORT+73), **73**, **RPT73M** (for MULTI-REPORT+73), and **73M** (for MULTI-73).
- **Prefix & Standard Callsign:** "Code" denotes prefix length; "call:std" is a 28-bit standard callsign (Section IV-A).
- **Non-Standard Callsigns & Suffixes:** "call:*n*" is an *n*-character non-standard callsign packed via Base-38 (Section IV-C); "Suf" encodes 1-bit portable modifiers (`0`=*none*, `1`=`/P`). Type 5 omits suffixes to pack two standard callsigns, locator, and SNR into 77 bits ($1+28+28+15+5=77$b).
- **Callsign Hashes:** "Hash" is a 24-bit CRC-24/Q hash ($2^{24} \approx 16.78 \times 10^6$ bins); "hash$^{\dagger}$" is the 16-bit DX hash (Section IV-D).
- **Multi-Station Fields:** Type 11 Fox pileup sequences [$\text{Prefix}_3$ | $\text{DX}_{16}$ | $\text{Tgt1}_{24}$ | $\text{SNR1}_5$ | $\text{Tgt2}_{24}$ | $\text{SNR2}_5$] (77 bits); Type 12 formats [$\text{Prefix}_8$ | $\text{DX}_{16}$ | $\text{Tgt1}_{24}$ | $\text{Tgt2}_{24}$ | $00000_2$] (77 bits, 5 zero-padded bits).
- **Modifiers, Locators & SNR:** "Mod" is a 20-bit modifier (Section IV-F); "loc" is a 15-bit locator (Section IV-E); "SNR" is a 5-bit report (Section IV-G).

### C. Protocol Boundary Conditions & Canonical Precedence

To eliminate ambiguity across station profiles, LQ formalizes eight canonical precedence and boundary rules:

1. **Strict Canonical Precedence:** Transmitters **must** use the most compact format fitting caller and target: for CALL, Type 5 requires both stations standard without suffix; Type 6 applies if either has `/P` or target is non-standard; Type 7 if caller is non-standard. For replies, Type 8 precedes Type 11 and Type 9 precedes Types 10/12 if both are standard.
2. **Deterministic Hash Disambiguation:** In Type 6, target hash matches against unsuffixed standard stations are rejected only if caller also lacks a suffix (Type 5 was mandatory); suffixed callers and Type 7 validly target standard stations. In 1-on-1 contacts, standard stations ignore hashed completions (Types 10, 11) from standard peers.
3. **Universal Locator and SNR in Type 6:** Type 6 allocates 4 prefix bits and 24 target hash bits, leaving 49 bits for caller callsign, suffix, 15-bit locator, and 5-bit SNR ($4+24+28+1+15+5=77$b). Suffixed callers deliver locator and SNR on their initial call without extra steps.
4. **Clear-Text Identity in Type 7:** Type 7 encodes the target via 20-bit hash ($H_{20} = H_{24} \gg 4$), the non-standard caller via clear-text 48-bit Base-38 representation ($\leq 9$ characters), 1-bit suffix (`/P`), and 5-bit SNR ($3+20+48+1+5=77$b), delivering unhashed identity directly in Step 2.
5. **Single-Target Multi-Station Framing:** When confirming one station in Fox mode (Types 11/12), Target 2 duplicates Target 1 (identical hash, and SNR in Type 11), decoded as a single station. In 1-on-1 contacts, Type 8 strictly supersedes Type 11; single-target Type 11 is reserved for non-standard callers or split F/H schedules.
6. **Collision Resistance:** Uniform 24-bit callsign hashes ($2^{24} \approx 16.78 \times 10^6$ bins) yield $P_{\text{collision}} \leq 0.75\%$ across 500 stations, disambiguated via local callsign caches and canonical precedence.
7. **Compound Callsigns:** Stations with callsigns exceeding 9 characters (up to 13 characters, e.g., `3B9/HB9IPH/P`, Category D) utilize the 69-bit Base-38 field in CQ frames (Type 4); directed replies use CQ or free text (Type 13).
8. **Verification & Compatibility:** Bit-exact codec traces and golden vectors across all 13 message formats are verified in the reference suite [11] across AWGN, Rayleigh fading, and clock-drift conditions. Compliant decoders treat unassigned opcodes (Types 14–16) as valid framing boundaries, safely ignoring them.

**Table 2:** Operational categories and corresponding 4-step message sequences.

| Cat. | Callsign / Mode Configuration | Step 1: CQ | Step 2: CALL | Step 3: REPORT+73 | Step 4: 73 |
|---|---|---|---|---|---|
| **A** | Standard (no suffixes) | Type 1 | Type 5 | Type 8 | Type 9 |
| **B** | Standard + std suffix (`/P`) | Type 1 | Type 6 | Type 8 | Type 9 |
| **C** | Non-standard ($\leq 9$ chars) + optional suffix | Type 2 / 3 | Type 7* / Type 6 | Type 11 | Type 10 |
| **D** | Long compound (10–13 chars) + optional suffix | Type 4 | —‡ | Type 11 | Type 12 |
| **E** | Multi-station reply | — | — | Type 11 (Fox) | Type 9 / 10 (Hounds)§ |

*In Step 2, Category C callers transmit Type 7 (clear-text callsign, suffix, SNR to target 20-bit hash); standard callers answering a Category C CQ transmit Type 6 (locator, SNR to target 24-bit hash). ‡Category D compound stations (10–13 chars) initiate contact via CQ (Type 4) or free text (Type 13). §A Hound with a 10–13-character callsign sends Type 12; the Fox may send an optional MULTI-73 (Type 12) after the exchange.

**Table 3:** Complete LQ message-type catalogue.

| Huffman Code | ID | Type | Bits (Payload) | | | | | | | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Code | Target | Suf | Caller | Suf | Modifier | Locator | SNR | |
| `0000000000` | 1 | CQ std+suf | 10 | — | — | 28 call:std | 1 suf | 20 mod | 15 loc | — | 74 |
| `0000000001` | 2 | CQ non-std 1 | 10 | — | — | 48 call:9 | 1 suf | — | 15 loc | — | 74 |
| `00000001` | 3 | CQ non-std 2 | 8 | — | — | 48 call:9 | 1 suf | 20 mod | — | — | 77 |
| `0000001` | 4 | CQ non-std 3 | 7 | — | — | 69 call:13 | 1 suf | — | — | — | 77 |
| `1` | 5 | CALL std (nosuf) | 1 | 28 call:std | —* | 28 call:std | —* | — | 15 loc | 5 SNR | 77 |
| `0100` | 6 | CALL std+suf | 4 | 24 hash | — | 28 call:std | 1 suf | — | 15 loc | 5 SNR | 77 |
| `001` | 7 | CALL non-std | 3 | 20 hash | — | 48 call:9 | 1 suf | — | — | 5 SNR | 77 |
| `0000000010` | 8 | REPORT+73 std+suf (RPT73) | 10 | 28 call:std | 1 suf | 28 call:std | 1 suf | — | — | 5 SNR | 73 |
| `0000000011` | 9 | 73 std+suf | 10 | 28 call:std | 1 suf | 28 call:std | 1 suf | — | — | — | 68 |
| `0001` | 10 | 73 non-std | 4 | 24 hash | — | 48 call:9 | 1 suf | — | — | — | 77 |
| `011` | 11 | MULTI-REPORT+73 (RPT73M) | 3 | 16 hash† + 2 × (24 hash + 5 SNR) | | | | | | | 77 |
| `00000111` | 12 | MULTI-73 (73M) | 8 | 16 hash† + 2 × 24 hash | | | | | | | 72 |
| `0101` | 13 | FREE TEXT | 4 | 73 text (Varicode) | | | | | | | 77 |
| `00001` | 14 | *(reserved A)* | 5 | 72 payload | | | | | | | 77 |
| `0000010` | 15 | *(reserved B)* | 7 | 70 payload | | | | | | | 77 |
| `00000110` | 16 | *(reserved C)* | 8 | 69 payload | | | | | | | 77 |

*Type 5 omits suffix fields to pack two standard callsigns, locator, and SNR into 77 bits ($1+28+28+15+5=77$b). †16-bit DX hash (Section IV-D). The "Total" column indicates active information bits; formats with Total $<77$ bits (Types 1, 2, 8, 9, 12) are right-padded with trailing zeros ($77-$ Total bits) to complete the 77-bit physical frame boundary prior to CRC-14 computation.

**Table 4:** LQ Varicode assignments for the 104-character free-text alphabet.

| Char | Varicode | Len | Freq | Char | Varicode | Len | Freq | Char | Varicode | Len | Freq | Char | Varicode | Len | Freq |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ␣ | `001` | 3 | 16.73% | 1 | `0000001011` | 10 | 0.09% | * | `000000000110` | 13 | 0.01% | Å | `00000000000010111` | 17 | <0.01% |
| E | `010` | 3 | 10.23% | Q | `0000001100` | 10 | 0.07% | [ | `000000000111` | 14 | 0.01% | Æ | `00000000000011000` | 17 | <0.01% |
| T | `0110` | 4 | 7.44% | ! | `0000001101` | 10 | 0.07% | ] | `000000000011111` | 14 | 0.01% | Ç | `00000000000011001` | 17 | <0.01% |
| A | `0111` | 4 | 6.59% | ? | `0000001110` | 10 | 0.07% | \| | `000000000000100` | 14 | 0.01% | È | `00000000000011010` | 17 | <0.01% |
| O | `1000` | 4 | 6.24% | Z | `0000001111` | 10 | 0.06% | / | `00000000000010100` | 15 | <0.01% | É | `00000000000011011` | 17 | <0.01% |
| N | `1001` | 4 | 5.96% | , | `000000001000` | 11 | 0.06% | $ | `000000000000010110` | 16 | <0.01% | Ê | `00000000000011100` | 17 | <0.01% |
| I | `1010` | 4 | 5.90% | . | `000000001001` | 11 | 0.05% | + | `000000000000010111` | 16 | <0.01% | Ë | `00000000000011101` | 17 | <0.01% |
| S | `1011` | 4 | 5.41% | 0 | `000000001010` | 11 | 0.05% | \n | `000000000000011000` | 16 | <0.01% | Ì | `00000000000011110` | 17 | <0.01% |
| R | `1100` | 4 | 5.00% | 2 | `000000001011` | 11 | 0.05% | % | `000000000000011001` | 16 | <0.01% | Í | `00000000000011111` | 17 | <0.01% |
| H | `1101` | 4 | 4.76% | 8 | `000000001100` | 11 | 0.04% | & | `000000000000011010` | 16 | <0.01% | Î | `00000000000001000` | 17 | <0.01% |
| D | `11100` | 5 | 3.48% | 3 | `000000001101` | 11 | 0.04% | < | `000000000000011011` | 16 | <0.01% | Ï | `00000000000001001` | 17 | <0.01% |
| L | `11101` | 5 | 3.21% | 4 | `000000001110` | 11 | 0.04% | > | `000000000000111000` | 17 | <0.01% | Đ | `00000000000001010` | 17 | <0.01% |
| C | `11110` | 5 | 2.34% | 5 | `000000001111` | 11 | 0.04% | @ | `000000000000111001` | 17 | <0.01% | Ñ | `00000000000001011` | 17 | <0.01% |
| U | `111110` | 6 | 2.24% | 9 | `000000000100` | 11 | 0.03% | \ | `000000000000111010` | 17 | <0.01% | Ò | `00000000000001100` | 17 | <0.01% |
| M | `111111` | 6 | 2.05% | 6 | `000000000101` | 11 | 0.03% | ^ | `000000000000111011` | 17 | <0.01% | Ó | `00000000000001101` | 17 | <0.01% |
| F | `000100` | 6 | 1.95% | 7 | `000000001100` | 12 | 0.03% | ` | `000000000000111100` | 17 | <0.01% | Ô | `00000000000001110` | 17 | <0.01% |
| W | `000101` | 6 | 1.63% | " | `000000001101` | 12 | 0.03% | { | `000000000000111101` | 17 | <0.01% | Õ | `00000000000001111` | 17 | <0.01% |
| P | `000110` | 6 | 1.60% | - | `000000001110` | 12 | 0.03% | } | `000000000000111110` | 17 | <0.01% | Ö | `00000000000000100` | 17 | <0.01% |
| G | `000111` | 6 | 1.56% | ' | `000000001111` | 12 | 0.02% | ~ | `000000000000111111` | 17 | <0.01% | Ø | `00000000000000101` | 17 | <0.01% |
| Y | `000010` | 6 | 1.46% | _ | `000000000100` | 12 | 0.02% | ¡ | `000000000000010000` | 17 | <0.01% | Ù | `00000000000000110` | 17 | <0.01% |
| B | `000011` | 6 | 1.18% | ; | `000000000101` | 12 | 0.02% | ¿ | `000000000000010001` | 17 | <0.01% | Ú | `00000000000000111` | 17 | <0.01% |
| V | `000010` | 7 | 0.85% | : | `000000000110` | 12 | 0.02% | À | `000000000000010010` | 17 | <0.01% | Û | `00000000000000010` | 17 | <0.01% |
| K | `00000110` | 8 | 0.53% | = | `0000000001110` | 13 | 0.02% | Á | `000000000000010011` | 17 | <0.01% | Ü | `00000000000000011` | 17 | <0.01% |
| ; | `00000111` | 8 | 0.34% | ( | `0000000001111` | 13 | 0.01% | Â | `000000000000010100` | 17 | <0.01% | Ý | `00000000000000001` | 17 | <0.01% |
| X | `000000100` | 9 | 0.16% | ) | `0000000000100` | 13 | 0.01% | Ã | `000000000000010101` | 17 | <0.01% | Þ | `00000000000000000001` | 18 | <0.01% |
| J | `0000001010` | 10 | 0.10% | # | `0000000000101` | 13 | 0.01% | Ä | `000000000000010110` | 17 | <0.01% | [FILL] | `00000000000000000000` | 18 | <0.01% |

## IV. FIELD ENCODINGS

To maintain compatibility, LQ adopts core field structures from FT8 [1, 2] (K9AN/K1JT)—28-bit callsigns, 15-bit grid locators, 20-bit CQ modifiers, and 5-bit signal reports—while introducing 1-bit portable suffixes, Base-38 non-standard callsigns, CRC-24/Q hashes, and a 104-character free-text Varicode.

### A. Standard Callsign (28 bits)

Adopted from FT8 [1, 2], standard ITU amateur callsigns canonicalize to 6 characters anchored around decimal digit $d$ at position 3: $c_1c_2ds_1s_2s_3$. Prefix $(c_1,c_2)$ right-aligns with space padding in $c_1$ if 1 character; suffix $(s_1,s_2,s_3)$ left-aligns with spaces. Callsigns matching $c_1c_2ds_1s_2s_3$ without slashes are standard; all others are non-standard (Section IV-C).

Index mappings: $c_1 \in \{␣,0\text{–}9,\text{A–Z}\}$ (radix 37: space $\to 0$, `0-9` $\to 1\ldots10$, `A-Z` $\to 11\ldots36$); $c_2 \in \{0\text{–}9,\text{A–Z}\}$ (radix 36: `0-9` $\to 0\ldots9$, `A-Z` $\to 10\ldots35$); $d \in \{0\text{–}9\}$ (radix 10); and $s_1,s_2,s_3 \in \{␣,\text{A–Z}\}$ (radix 27: space $\to 0$, `A-Z` $\to 1\ldots26$). With indices $(n_1,\ldots,n_6)$, mixed-radix serialization packs into 28-bit integer $N_{28}$:

$$N_{28} = (n_1\cdot 36\cdot 10 + n_2\cdot 10 + n_3)\cdot 27^3 + n_4\cdot 27^2 + n_5\cdot 27 + n_6. \tag{3}$$

Standard callsigns occupy $0 \le N_{28} \le N_{\max} = 262{,}177{,}559 < 2^{28}$; subsequent states encode tokens `DE` = 262,177,560, `QRZ` = 262,177,561, and `CQ` = 262,177,562. Decoders validate radix ranges to reject corrupted values.

### B. Standard Dedicated Suffixes (1 bit)

To support field activations (POTA/SOTA) without hashing, frames reserve an explicit 1-bit suffix: `0`=*none* (e.g., `HB9IPH`), `1`=`/P` (portable).

### C. Non-Standard Callsigns (Base-38 Big-Integer Packing)

Non-standard or slashed callsigns pack as Base-38 strings over alphabet ␣ (0), `A–Z` (1–26), `/` (27), and `0–9` (28–37), right-padded with spaces to length $L$ and evaluated big-endian as $N_{\text{b38}} = \sum_{i=0}^{L-1} v_i \cdot 38^{L-1-i}$. **9-Char Callsigns (48 bits):** $38^9 < 2^{48}$ (Types 2, 3, 7, 10), supporting up to 9 characters plus 1-bit portable suffix (`/P`, e.g., `EA6/HB9IP/P`) or slashed calls (`HB9IPH/Q`). **13-Char Callsigns (69 bits):** $38^{13} < 2^{69}$ (Type 4), supporting compound callsigns up to 13 characters plus suffix in CQ (`3B9/HB9IPH/P`). Unnormalized whitespace is rejected.

**Alphabet Disambiguation:** This Base-38 alphabet (space $\to 0$, `A-Z` $\to 1\ldots26$, `/` $\to 27$, `0-9` $\to 28\ldots37$) places letters before digits, unlike the legacy WSJT-X order (`0-9` $\to 1\ldots10$, `A-Z` $\to 11\ldots36$) used in the 16-bit hash (Section IV-D).

### D. Callsign Hashes (24-bit, 20-bit, and 16-bit) & Dehashing

Non-standard stations addressed in directed frames are identified via collision-resistant hashes: **24-Bit Callsign Hash ($H_{24}$):** Computed over trimmed ASCII callsigns via CRC-24/Q [9] ($G(x)$ = `0x1864CFB`, init/final `0x000000`, non-reflected): $H_{24}(\text{call})$ = CRC-24/Q(call), yielding $2^{24}$ = 16,777,216 states ($P_{\text{collision}} \le 0.75\%$ across 500 stations). **20-Bit Target Hash ($H_{20}$):** Used in Type 7 directed replies as 20 MSBs: $H_{20}(\text{call}) = \lfloor H_{24}(\text{call})/2^4 \rfloor = H_{24}(\text{call}) \gg 4$, yielding $2^{20}$ = 1,048,576 states (e.g., $H_{20}$ = `0x01A4F` for $H_{24}$ = `0x01A4F2`), conserving 4 payload bits for reports while maintaining high collision resistance.

**16-Bit DX Callsign Hash ($H_{16}$):** Used in multi-station frames (Types 11 and 12), computed via the WSJT-X 16-bit multiplicative hash [2]. Callsigns right-pad to 11 characters over alphabet (␣ $\to 0$, `0-9` $\to 1\ldots10$, `A-Z` $\to 11\ldots36$, `/` $\to 37$):

$$H_{16}(\text{call}) = \lfloor (\textstyle\sum_{i=0}^{10} v_i \cdot 38^{10-i} \cdot 47055833459)/2^{48} \rfloor \bmod 2^{16}, \tag{4}$$

yielding 65,536 states (e.g., $H_{16}$(`HB9IPH`) = `0x7A8B`); to prevent collisions, decoders restrict $H_{16}$ dehashing to actively called stations (evicted upon QSO completion or timeout).

**UI Resolution:** Resolved hashes display bracketed by callsign (e.g., `<HB9IPH>`); unresolved hashes display in hex (e.g., `<01A4F2>`, `<01A4F>`, or `<7A8B>`).

### E. Maidenhead Grid Locator (15 bits)

Following the FT8 standard [1, 2, 8], four-character Maidenhead grid locators ($F_1F_2D_1D_2$, with fields $F_1,F_2 \in$ [A–R] and squares $D_1,D_2 \in$ [0–9]) divide the globe into $18\times18 = 324$ fields of $10\times10 = 100$ squares (32,400 total squares). Packed into 15 bits ($N_{\text{grid}} < 32{,}400 < 2^{15} = 32{,}768$):

$$\begin{aligned} N_{\text{grid}} = {} & (F_1 - \text{'A'})\cdot 1800 + (F_2 - \text{'A'})\cdot 100 \\ & + (D_1 - \text{'0'})\cdot 10 + (D_2 - \text{'0'}). \end{aligned} \tag{5}$$

Values $0 \le N_{\text{grid}} \le 32{,}399$ represent valid Maidenhead squares; $N_{\text{grid}} = 32{,}400$ is the blank locator sentinel; values $32{,}401 \le N_{\text{grid}} \le 32{,}767$ are rejected.

### F. CQ Modifiers (20 bits)

Directed CQ frames (Types 1 and 3) adopt the 20-bit modifier structure from WSJT-X / FT8 [1, 2] ($2^{20}$ = 1,048,576 states): *1) Unmodified CQ:* $N_{\text{mod}} = 0$ designates general CQ calls. *2) 3-Digit Numeric (000–999):* $1 \le N_{\text{mod}} \le 1000$ conveys numeric designators (e.g., net/channel IDs `CQ 040`), mapped as $N_{\text{mod}} = \text{val} + 1$. *3) Alphanumeric Tokens ($\le 4$ chars):* $1001 \le N_{\text{mod}} \le 1{,}048{,}575$ encodes tokens up to 4 characters (e.g., `DX`, `NA`, `POTA`, `SOTA`, `QRP`, `CQWW`; values $> 2^{20} - 1$ rejected). Tokens $< 4$ characters right-pad with spaces over a 32-symbol alphabet (␣ $\to 0$, `A-Z` $\to 1\ldots26$, `0-4` $\to 27\ldots31$):

$$N_{\text{mod}} = 1000 + \sum_{i=0}^{3} v_i \cdot 32^{3-i}. \tag{6}$$

Omitting digits 5–9 ensures $32^4 = 2^{20}$ fits without overflow; longer modifiers use 4-character abbreviations (`WPX`) or free text.

### G. Signal Reports (5 bits)

Following FT8 [1], signal reports convey measured SNR from $-26$ dB to $+5$ dB via integer $N_{\text{snr}} = \text{clamp}(\text{SNR}_{\text{dB}} + 26, 0, 31) \in [0, 31]$. In directed CALL replies (Types 5–7), reports display without prefix (e.g., `-03`); in acknowledgments (REPORT+73, Types 8 and 11), `R` is prepended (e.g., `R-03`).

### H. Free-Text Varicode Encoding (73 bits)

Unstructured text in Type 13 (FREE TEXT, 73 payload bits) uses a 104-character prefix code (Table 4) [7] optimized from character frequencies [10]. The alphabet spans uppercase `A-Z`, digits `0-9`, space, 32 punctuation symbols, 32 Latin-1 characters (¡–Þ), newline `\n`, ↵, and `[FILL]`. Codewords range from 3 to 18 bits (mean 4.25 bits/char), conveying **≈17.2 characters** on average (+32.3% over conventional 13-character payloads). Unused bits are zero-padded; decoders terminate extraction at the first zero bit without emitting fill tokens.

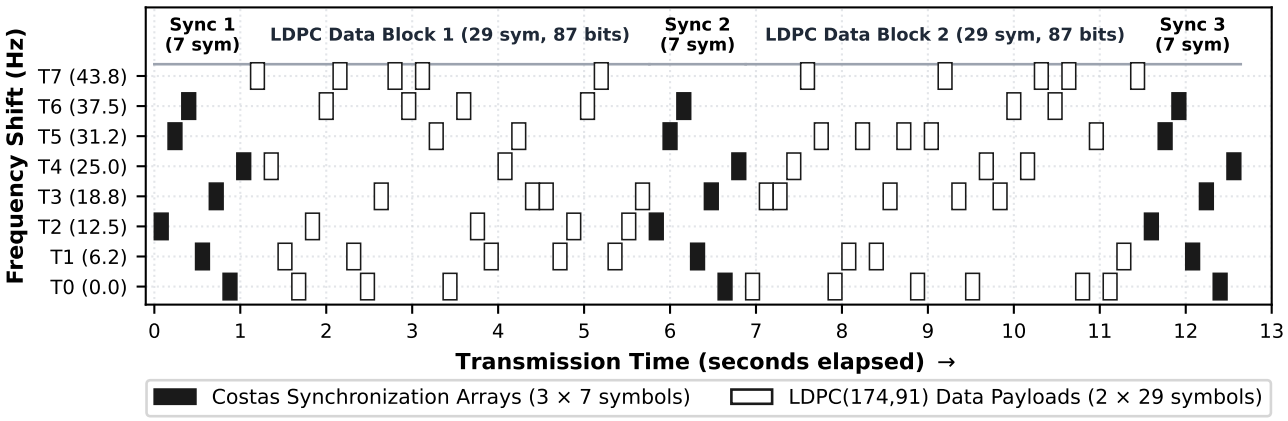



**Figure 1:** LQ8 physical frame structure (8-GFSK, 50.0 Hz bandwidth, 12.64 s duration).

## V. PHYSICAL LAYER

To ensure equal weak-signal sensitivity (−21.0 dB SNR in 2500 Hz bandwidth) and transceiver compatibility, LQ8 deliberately adopts the core physical-layer modulation and error-correction coding of the FT8 protocol [1], engineered by Steve Franke (K9AN) and Joe Taylor (K1JT).

### A. Channel Coding: CRC-14 and Systematic LDPC(174,91)

Link-layer encapsulation preserves FT8's channel-coding structure [1]: (1) **77-Bit Payload** from the application layer; (2) **14-Bit CRC Checksum** ($G_{\mathrm{CRC14}}(x)$, Table 5) evaluated by shifting the 77-bit payload MSB-first into a 14-bit LFSR using FT8's generator polynomial `0x2757` followed by 14 trailing zeros, appending remainder $(d_{77}\dots d_{90})$ to form a 91-bit systematic block ($P_{\mathrm{false}} \le 2^{-14} \approx 6.1\times10^{-5}$); 5 zero bits pad to a 12-byte boundary; and (3) **Systematic LDPC(174,91) FEC** utilizing the identical Gallager generator matrix $G_{83\times91}$ (Table 6) [1, 5] producing 83 parity bits ($R \approx 0.523$, 174 codeword bits).

Systematic codeword $\mathbf{c} \in \mathbb{F}_2^{174}$ sets $c_j = d_j$ for $0 \le j < 91$. Parity bits are:

$$c_{91+i} = \bigoplus_{j=0}^{90} G_{i,j}\cdot d_j, \quad 0 \le i < 83, \tag{7}$$

where $G_{83\times91}$ is the Gallager matrix in Table 6 (23 hex digits/row, 91 MSB-aligned bits; bit 0 of digit 23 is zero pad). CRC-14 parameters are listed in Table 5.

**Table 5:** CRC-14 channel coding parameter specification.

| Parameter | Value / Specification |
|---|---|
| Generator Polynomial $G_{\mathrm{CRC14}}(x)$ | $x^{14}+x^{13}+x^{10}+x^9+x^8+x^6+x^4+x^2+x+1$ |
| Hex / LFSR Representation | `0x2757` (14-bit: $10011101010111_2$; 15-bit monic: `0x6757`) |
| Initial State / Final XOR | `0x0000` (14 zeros) / `0x0000` (uninverted) |
| Input / Output Reflection | `False` / `False` (MSB-first evaluation & output) |
| Bit Framing & Padding | 77 payload bits + 14 CRC bits = 91 systematic bits (+5 pad → 96) |
| Undetected Error Probability | $P_{\mathrm{false}} \le 2^{-14} \approx 6.10\times10^{-5}$ |
| Invalid Checksum Sentinel | Remainder `0x0000` rejected as corrupt |
| Target FEC Input Block | Feeds LDPC(174,91) generator $G_{83\times91}$ |

### B. LQ8 Physical Transport & UTC Slot Framing

LQ8 adopts FT8's continuous-phase 8-GFSK transport parameters [1]: 79 symbols in 12.64 s within 15.0 s UTC slots ($R_s = 6.25$ Bd, $\Delta f = 6.25$ Hz, 50.0 Hz footprint, −21.0 dB SNR sensitivity, Figure 1). Transmissions occupy 3.0 kHz USB segments with multiple concurrent subcarrier offsets.

For timestamp $t_{\mathrm{UTC}}$, slot index and parity $s \in \{0,1\}$ evaluate as $k = \lfloor t_{\mathrm{UTC}}/T_{\mathrm{slot}} \rfloor$ and $s = k \bmod 2$ (Even :00/:30 vs. Odd :15/:45). Guard interval $T_{\mathrm{guard}} = T_{\mathrm{slot}} - T_{\mathrm{tx}}$ (2.36 s in LQ8, 2.46 s in LQ4) provides propagation and T/R turnaround margin; stations alternate parity across cycles.

### C. Modulation Mapping & Continuous-Phase Synthesis

The 3-bit Gray code maps channel bit triplets to tones: `000`→0, `001`→1, `011`→2, `010`→3, `110`→4, `100`→5, `101`→6, and `111`→7. Continuous-phase audio synthesis $s(t) = A(t)\cos(\phi(t))$ at $f_s = 12{,}000$ Hz employs half-symbol raised-cosine envelope shaping ($T_{\mathrm{ramp}} = T_s/2 = 80.0$ ms in LQ8, 160.0 ms in LQ16) and Gaussian frequency trajectories ($BT = 2.0$):

$$\phi(t) = \phi_0 + 2\pi \int_0^t \left( f_0 + \sum_{k=0}^{N_{\mathrm{sym}}-1} a_k \cdot \Delta f \cdot g(\tau - kT_s) \right) d\tau, \tag{8}$$

where tone index $a_k \in \{0,1,\dots,M-1\}$ is uncentered with tone 0 at base subcarrier $f_0$ ($\Delta f = 6.25$ Hz in LQ8). Frequency pulse $g(t)$ is truncated to duration $L = 3T_s$ ($t \in [-1.5T_s, +1.5T_s]$):

$$g(t) = \frac{1}{2T_s}\left[ \mathrm{erf}\left( 2\pi B \frac{t + T_s/2}{\sqrt{2\ln 2}} \right) - \mathrm{erf}\left( 2\pi B \frac{t - T_s/2}{\sqrt{2\ln 2}} \right) \right]. \tag{9}$$

### D. Costas Synchronization & DSP Pipeline

While LQ8 shares FT8's modulation, CRC-14, and LDPC codes, it deliberately departs in its synchronization structure to achieve physical cross-protocol isolation. Rather than FT8's sync sequence ($[3,1,4,0,6,5,2]$), 8-GFSK modes (LQ8 and LQ16) synchronize via three 7-symbol Costas arrays [6] ($C_7 = [2,5,6,1,3,0,4]$) at symbols 0–6, 36–42, and 72–78. The 174-bit systematic LDPC codeword is partitioned without bit interleaving into two 87-bit data blocks: Data Block 1 (symbols 7–35, $29\times3 = 87$ bits) carries codeword bits $c_0\dots c_{86}$, while Data Block 2 (symbols 43–71) carries $c_{87}\dots c_{173}$. Each symbol maps 3 consecutive bits MSB-first via the 3-bit Gray code. Preambles exhibit ideal autocorrelation with peak sidelobe $\le 1$ and $\le 1$ coincidence hit against FT8, preventing false decodes.

Receiver DSP executes the multi-pass decoding paradigm from WSJT-X [1, 2]: sliding STFT (Hann window, 4-hop oversampling) generating spectral density $S[t_i, f_j]$, 2D matched filter detection across ±0.5 s timing windows, 3-point parabolic frequency refinement ($|\Delta f_{\mathrm{err}}| < 0.1$ Hz), soft LLR extraction, log-domain LDPC belief-propagation decoding (up to 25 passes, terminating early upon $\mathbf{H}\cdot\hat{\mathbf{c}}^T = \mathbf{0}$ and CRC-14 verification), and time-frequency candidate subtraction to decode weaker signals.

**Table 6:** The complete Gallager LDPC(174,91) generator matrix $G_{83\times91}$ (83 rows × 23 hex digits). Each row represents 92 bits MSB-aligned; bit 0 of the 23rd hex digit is an unused zero padding bit.

| Row | Generator Vector (23 Hex Digits) | Row | Generator Vector (23 Hex Digits) | Row | Generator Vector (23 Hex Digits) |
|---|---|---|---|---|---|
| 00 | 8329ce11bf31eaf509f27fc | 28 | 8eba1a13db3390bd6718cec | 56 | 7ee18230c583cccc57d4b08 |
| 01 | 761c264e25c259335493132 | 29 | 753844673a27782cc42012e | 57 | a066cb2fedafc9f52664126 |
| 02 | dc265902fb277c6410a1bdc | 30 | 06ff83a145c37035a5c1268 | 58 | bb23725abc47cc5f4cc4cd2 |
| 03 | 1b3f417858cd2dd33ec7f62 | 31 | 3b37417858cc2dd33ec3f62 | 59 | ded9dba3bee40c59b5609b4 |
| 04 | 09fda4fee04195fd034783a | 32 | 9a4a5a28ee17ca9c324842c | 60 | d9a7016ac653e6decdc9036 |
| 05 | 077cccc11b8873ed5c3d48a | 33 | bc29f465309c977e89610a4 | 61 | 9ad46aed5f707f280ab5fc4 |
| 06 | 29b62afe3ca036f4fe1a9da | 34 | 2663ae6ddf8b5ce2bb29488 | 62 | e5921c77822587316d7d3c2 |
| 07 | 6054faf5f35d96d3b0c8c3e | 35 | 46f231efe457034c1814418 | 63 | 4f14da8242a8b86dca73352 |
| 08 | e20798e4310eed27884ae90 | 36 | 3fb2ce85abe9b0c72e06fbe | 64 | 8b8b507ad467d4441df770e |
| 09 | 775c9c08e80e26ddae56318 | 37 | de87481f282c153971a0a2e | 65 | 22831c9cf1169467ad04b68 |
| 10 | b0b811028c2bf997213487c | 38 | fcd7ccf23c69fa99bba1412 | 66 | 213b838fe2ae54c38ee7180 |
| 11 | 18a0c9231fc60adf5c5ea32 | 39 | f0261447e9490ca8e474cec | 67 | 5d926b6dd71f085181a4e12 |
| 12 | 76471e8302a0721e01b12b8 | 40 | 4410115818196f95cdd7012 | 68 | 66ab79d4b29ee6e69509e56 |
| 13 | ffbccb80ca8341fafb47b2e | 41 | 088fc31df4bfbde2a4eafb4 | 69 | 958148682d748a38dd68baa |
| 14 | 66a72a158f9325a2bf67170 | 42 | b8fef1b6307729fb0a078c0 | 70 | b8ce020cf069c32a723ab14 |
| 15 | c4243689fe85b1c51363a18 | 43 | 5afea7acccb77bbc9d99a90 | 71 | f4331d6d461607e95752746 |
| 16 | 0dff739414d1a1b34b1c270 | 44 | 49a7016ac653f65ecdc9076 | 72 | 6da23ba424b9596133cf9c8 |
| 17 | 15b48830636c8b99894972e | 45 | 1944d085be4e7da8d6cc7d0 | 73 | a636bcbc7b30c5fbeae67fe |
| 18 | 29a89c0d3de81d665489b0e | 46 | 251f62adc4032f0ee714002 | 74 | 5cb0d86a07df654a9089a20 |
| 19 | 4f126f37fa51cbe61bd6b94 | 47 | 56471f8702a0721e00b12b8 | 75 | f11f106848780fc9ecdd80a |
| 20 | 99c47239d0d97d3c84e0940 | 48 | 2b8e4923f2dd51e2d537fa0 | 76 | 1fbb5364fb8d2c9d730d5ba |
| 21 | 1919b75119765621bb4f1e8 | 49 | 6b550a40a66f4755de95c26 | 77 | fcb86bc70a50c9d02a5d034 |
| 22 | 09db12d731faee0b86df6b8 | 50 | a18ad28d4e27fe92a4f6c84 | 78 | a534433029eac15f322e34c |
| 23 | 488fc33df43fbdeea4eafb4 | 51 | 10c2e586388cb82a3d80758 | 79 | c989d9c7c3d3b8c55d75130 |
| 24 | 827423ee40b675f756eb5fe | 52 | ef34a41817ee02133db2eb0 | 80 | 7bb38b2f0186d46643ae962 |
| 25 | abe197c484cb74757144a9a | 53 | 7e9c0c54325a9c15836e000 | 81 | 2644ebadeb44b9467d1f42c |
| 26 | 2b500e4bc0ec5a6d2bdbdd0 | 54 | 3693e572d1fde4cdf079e86 | 82 | 608cc857594bfbb55d69600 |
| 27 | c474aa53d70218761669360 | 55 | bfb2cec5abe1b0c72e07fbe | | |

## VI. EXTENDED MODES: LQ16, LQ4, AND LQ2

While **LQ8** is the primary mode for 15-second scheduled weak-signal communications, the 77-bit variable-length architecture scales across three complementary physical profiles (Table 7):

- **LQ16 (Slow / Ultra Weak):** 8-GFSK mode for severe weak-signal conditions (−24.0 dB SNR threshold in 2500 Hz) with 320 ms symbol period (25.0 Hz bandwidth, $BT = 2.0$, 25.28 s duration) for extreme weak-signal and EME paths.
- **LQ4 (Fast / Contest):** 4-GFSK mode for rapid contesting and pileups (30.0 s contact, 120 QSOs/h standard, 480 QSOs/h Fox/Hound) with 48 ms symbol period, 83.3 Hz bandwidth, and 5.04 s transmission in 7.5 s UTC slots.
- **LQ2 (Turbo Burst):** 4-GFSK high-throughput profile for dynamic burst channels (15.0 s full contact, 240 QSOs/h standard, 960 QSOs/h Fox/Hound) with 24 ms symbol period, 166.7 Hz bandwidth, and 2.52 s transmission in 3.75 s slots for meteor scatter.

### A. 4-GFSK Frame Structure & Modulation

LQ4 and LQ2 transmit 105 symbols/frame using continuous-phase 4-GFSK ($BT = 1.0$, tone spacing $\Delta f = 1/T_s$, modulation index $h = 1.0$):

- **Consolidated Framing Sequence:** The 105-symbol frame is partitioned into: tone-0 ramp-up (sym 0, $T_{\text{ramp}} = T_s/2$, 24 ms in LQ4, 12 ms in LQ2); Costas sync $S_1 = [0,2,3,1]$ (syms 1–4); Data Block 1 (syms 5–33, 29 symbols, codeword bits 0…57); Costas $S_2 = [1,3,2,0]$ (syms 34–37); Data Block 2 (syms 38–66, 29 symbols, bits 58…115); Costas $S_3 = [2,0,1,3]$ (syms 67–70); Data Block 3 (syms 71–99, 29 symbols, bits 116…173); Costas $S_4 = [3,1,0,2]$ (syms 100–103); and tone-0 ramp-down (sym 104, $T_{\text{ramp}} = T_s/2$).
- **Scrambling & Whitening:** Prior to CRC-14/LDPC channel encoding, the 77-bit payload is bitwise XORed with the 10-byte whitening sequence (`4A 5E 89 B4 B0 8A 79 55 BE 28`$_{16}$) to suppress discrete spectral lines. Bits 77–79 (byte 9, `0x28`) are zeros prior to CRC-14 calculation.
- **Data Framing & Gray Coding:** Codeword bits are mapped sequentially without bit interleaving. The three 29-symbol data blocks map consecutive pairs $(b_{2k}, b_{2k+1})$ of the 174-bit codeword to 4-GFSK tones via Gray coding (`00`→0, `01`→1, `11`→2, `10`→3).
- **Ramping & Guard Margins:** Tone-0 symbols at indices 0 and 104 frame the sequence (5.04 s for LQ4, 2.52 s for LQ2). Guard intervals (2.46 s in LQ4, 1.23 s in LQ2) absorb propagation delay and T/R turnaround.

### B. Profile Selection and Channel Matching

Transceivers dynamically select operational profiles according to ionospheric channel coherence time $T_c$ and link signal-to-noise ratio:

- When SNR $< -21.0$ dB and Doppler spread $\sigma_f \leq 0.5$ Hz, LQ16 provides a +3.0 dB gain over LQ8, sustaining readability down to −24.0 dB for extreme paths.
- In rapid-fading channels ($T_c < 10$ s), LQ4's 48 ms symbols and 7.5 s slots ensure robust sync, completing verified contacts in 30.0 s before phase degrades.
- For dynamic burst channels (e.g., meteor scatter), LQ2's 2.52 s bursts capture transient trails, completing 4-slot exchanges in 15.0 s.

**Table 7:** Multi-mode parameter and performance comparison across standard reference modes and the LQ family.

| Parameter | FT8 (Ref) | FT4 (Ref) | LQ8 | LQ16 | LQ4 | LQ2 |
|---|---|---|---|---|---|---|
| Modulation | 8-GFSK | 4-GFSK | 8-GFSK | 8-GFSK | 4-GFSK | 4-GFSK |
| Bandwidth | 50.0 Hz | 83.3 Hz | 50.0 Hz | 25.0 Hz | 83.3 Hz | 166.7 Hz |
| Time slot | 15.0 s | 7.5 s | 15.0 s | 30.0 s | 7.5 s | 3.75 s |
| Transmission time | 12.64 s | 5.04 s | 12.64 s | 25.28 s | 5.04 s | 2.52 s |
| Sensitivity (SNR) | −21.0 dB | −17.5 dB | −21.0 dB | −24.0 dB | −17.5 dB | −14.0 dB |
| Relative RF power req. | **0 dB (ref)** | +3.5 dB (2.2×) | 0 dB (ident.) | −3.0 dB (0.5×) | +3.5 dB (2.2×) | +7.0 dB (5.0×) |
| Type coding | Fixed | Fixed | Huffman | Huffman | Huffman | Huffman |
| QSO messages | 5–6 | 5–6 | 4 | 4 | 4 | 4 |
| QSO duration | 75.0–90.0 s | 37.5–45.0 s | 60.0 s | 120.0 s | 30.0 s | 15.0 s |
| Standard QSO Rate* | 40–48/h | 80–96/h | 60/h | 30/h | 120/h | 240/h |
| Pileup Rate (1-carrier)* | 60/h | 120/h | 240/h | 120/h | 480/h | 960/h |
| Pileup Rate (2-carrier)* | 120/h | 240/h | 480/h | 240/h | 960/h | 1920/h |
| Standard Suffixes | /P, /R | /P, /R | /P | /P | /P | /P |
| Max non-std callsign | 11 chars | 11 chars | 13 + std suffix | 13 + std suffix | 13 + std suffix | 13 + std suffix |
| Callsign hash length | 10/12/22-b | 10/12/22-b | 24/20/16-b | 24/20/16-b | 24/20/16-b | 24/20/16-b |
| Free-text alphabet | 42 chars | 42 chars | 104 chars | 104 chars | 104 chars | 104 chars |
| Free-text capacity | 71 bits (∼13 c) | 71 bits (∼13 c) | 73 bits (∼17 c) | 73 bits (∼17 c) | 73 bits (∼17 c) | 73 bits (∼17 c) |

*FT8/FT4 contacts require 5–6 messages (75–90 s / 37.5–45 s) using `RR73` or `RRR`/73; LQ completes contacts in 4 messages (60 s in LQ8). Peak rates assume continuous zero-loss operation. Pipelined pileup rates reflect F/H mode confirming 1 station/cycle in FT8/FT4 vs. 2 in LQ (Type 11).

## VII. CONCLUSION

**LQ8** demonstrates that continuous-phase 8-GFSK and systematic LDPC$(174, 91)$ transport deliver higher throughput through variable-length prefix framing while maintaining full backward physical compatibility and −21.0 dB SNR sensitivity in a 2500 Hz reference bandwidth. By optimizing message prefix allocations to match operational entropy, LQ8 packs callsigns, locators, and reports into a 77-bit budget, finalizing confirmed two-way contacts in 4 transmission slots (60 s). For high-density pileups, LQ8 introduces the single-carrier MULTI-REPORT+73 protocol, confirming two answering stations simultaneously without RF power-splitting penalties and achieving throughputs up to 240 QSOs/h (480 QSOs/h in dual-carrier operation).

The unified protocol family extends this architecture across three complementary physical profiles. LQ16 scales symbol duration to 320 ms, delivering a +3.0 dB sensitivity advantage (−24.0 dB SNR threshold) across ultra-weak fading and EME paths. For rapid contesting and dynamic propagation, LQ4 (7.5 s slots) and LQ2 (3.75 s slots) utilize 4-GFSK modulation and Costas sync arrays to complete contacts in 30.0 s and 15.0 s, supporting burst rates up to 960 QSOs/h on VHF/UHF meteor scatter and sporadic-E openings.

The complete software suite is implemented in the open-source C++ reference library (`lq_lib`) under the MIT License, featuring an automated verification harness achieving $> 90\%$ line coverage alongside comprehensive golden vector test suites across all operational profiles. End-to-end field verification has been demonstrated in real-world portable activations via **qFT8** across HF bands. Bit-exact codec traces for all 13 message formats are verified under AWGN, Rayleigh fading, and clock drift; reference sensitivities span −14.0 dB (LQ2) to −24.0 dB (LQ16).

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